SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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CSA37- SOFTWARE TESTING

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Requirement Analysis and Positive Test Scenarios and Test Cases for an Online Examination System

Aim

To study the requirements of an Online Examination System, and to prepare positive test scenarios and test cases that confirm the application works correctly for valid inputs and normal user actions.

Problem Statement

An online examination system lets an examiner set up exams and lets candidates write them on a computer, with automatic marking. Many candidates use it at the same time, and the results affect their careers, so a defect can cause wrong marks, lost answers or unfair exams.

Testing is needed to check that the application:

- Candidates can log in, write an exam and get correct results.
- Invalid inputs, cheating attempts and unauthorized access are handled safely.
- Timing, submission and evaluation rules follow the requirements.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	Candidates log in with a register number and password. Examiners log in through a separate admin login.
FR2	The admin can create an exam with questions, marks, duration and a scheduled start and end time.
FR3	A candidate can start the exam only within the scheduled time window.
FR4	A countdown timer shows the remaining time, and the exam is submitted automatically when time ends.
FR5	Questions are shown one per page. The candidate can move to the next or previous question and mark questions for review.
FR6	Selected answers are saved automatically, and a candidate can change an answer before submitting.
FR7	The exam can be submitted only once. A confirmation is asked before the final submission.
FR8	Evaluation is automatic. One mark is given for each correct answer and 0.25 mark is deducted for each wrong answer.
FR9	After 3 warnings for switching tabs or windows, the exam is submitted automatically.
FR10	If the network breaks, the candidate can log in again and continue with the remaining time and saved answers.

ID	Requirement
FR11	Results are shown to candidates only after the admin publishes them.
FR12	The admin can view the result report. A candidate cannot open admin pages, and one account cannot be logged in on two devices at once.

2 Software and environment requirements

Item	Details
Application	Web application (Node.js / Java back end, React front end)
Database	MySQL (users, exams, questions, answers, results)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, JMeter, bug tracker (Jira)
Test data	Question bank of 20 MCQs, sample candidates and admin accounts, an exam scheduled for a fixed time window

Positive Test Scenarios

A positive test scenario checks that a feature works as expected when the user gives valid input and follows the normal flow.

Scenario ID	Module	Test Scenario	Requirement
OE-PS01	Login	Candidate logs in with valid credentials	FR1
OE-PS02	Exam setup	Admin creates an exam	FR2
OE-PS03	Start exam	Start the exam inside the time window	FR3
OE-PS04	Timer	Timer counts down	FR4
OE-PS05	Navigation	Move between questions and mark for review	FR5
OE-PS06	Answers	Answer is saved automatically	FR6
OE-PS07	Answers	Change an answer before submission	FR6
OE-PS08	Submission	Submit the exam after confirmation	FR7
OE-PS09	Timer	Automatic submission when time ends	FR4, FR7
OE-PS10	Evaluation	Score is calculated correctly	FR8
OE-PS11	Recovery	Resume after a network interruption	FR10
OE-PS12	Results	Admin publishes results and candidates view them	FR11, FR12

Test Cases

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
OE-PT01	OE-PS01	1. Open the candidate login page 2. Enter register number and password 3. Click Login	192312070; Exam@123	Candidate is logged in and the list of scheduled exams is shown.
OE-PT02	OE-PS02	1. Log in as admin 2. Open Create Exam 3. Enter title, questions, marks, duration and schedule 4. Save	20 MCQs; 30 minutes; 10:00 to 12:00	Exam is created and appears in the scheduled exam list.
OE-PT03	OE-PS03	1. Log in at 10:05 2. Open the scheduled exam 3. Click Start Exam	Current time inside 10:00 to 12:00	Exam starts and the first question is displayed.
OE-PT04	OE-PS04	1. Start the exam 2. Observe the timer for one minute	30-minute exam	Timer shows the remaining time and decreases every second.
OE-PT05	OE-PS05	1. Click Next 2. Click Previous 3. Click Mark for Review on question 2	Questions 1 to 3	Navigation works, and the marked question is highlighted in the palette.
OE-PT06	OE-PS06	1. Select an option for question 1 2. Move to question 2 and back	Option B	The selected option is still selected and saved.
OE-PT07	OE-PS07	1. Select option B 2. Change it to option C	Option B then C	The latest option C is saved.
OE-PT08	OE-PS08	1. Click Submit 2. Click Yes on the confirmation	20 questions answered	Exam is submitted and a submission message is shown.
OE-PT09	OE-PS09	1. Keep the exam open until the timer reaches zero	Duration 30 minutes	Exam is submitted automatically with the saved answers.
OE-PT10	OE-PS10	1. Answer 15 correctly, 3 wrongly and leave 2 2. Submit 3. View the result	15 correct; 3 wrong; 2 skipped	Score is $15 - (3 \times 0.25) = 14.25$.
OE-PT11	OE-PS11	1. Start the exam 2. Turn off the network for 1 minute 3. Turn it on and log in again	Network off for 1 minute	The exam resumes with the saved answers and the correct remaining time.

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
OE-PT12	OE-PS12	1. Log in as admin 2. Click Publish Results 3. Log in as a candidate 4. Open Results	Exam closed	Candidate sees the score. The admin can view the full report.

Result / Conclusion

Result: 12 positive test scenarios and 12 test cases were prepared for the Online Examination System, covering 11 of the 12 functional requirements (no positive scenario for FR9). The test cases by module are:

Module	Test Scenarios	Test Cases
Login	1	1
Exam setup	1	1
Start exam	1	1
Timer	2	2
Navigation	1	1
Answers	2	2
Submission	1	1
Evaluation	1	1
Recovery	1	1
Results	1	1
Total	12	12

Conclusion: The positive test cases cover the main user flows of the Online Examination System, from the first action to the final result. When they are executed on a build, they confirm that the valid-input behaviour matches the specification. They should be run first in every test cycle and kept in the regression suite, with the negative test cases added to check invalid inputs.